

Data Type and Point Cloud

1. Introduction to Point Cloud

The things we observe with our eyes are three-dimensional, but in computer vision, using regular cameras only provides two-dimensional information about an object, which cannot achieve the same observation effect as human eyes. In order for computers to represent three-dimensional objects, computer vision needs to transition into three-dimensional vision, and point clouds are the way to express three-dimensional vision.

2. Understanding Point Cloud Data Types

Point cloud data refers to a collection of vectors in a three-dimensional coordinate system, typically expressed in the X, Y, and Z axes of the three-dimensional coordinate system. It is commonly used to represent the surface shape of an object. Point cloud data can also represent attributes such as RGB colors, grayscale values, depth, etc., of an object's surface.

Robot point cloud data is acquired through Lidar and depth cameras, both of which are capable of measurements.

Common formats for storing point cloud data include pts, asc, dat, stl, imw, xyz, txt, csv, and etc. Point cloud data, much like an attribute table, records the X, Y, Z positions of corresponding points along with various attributes associated with each point.

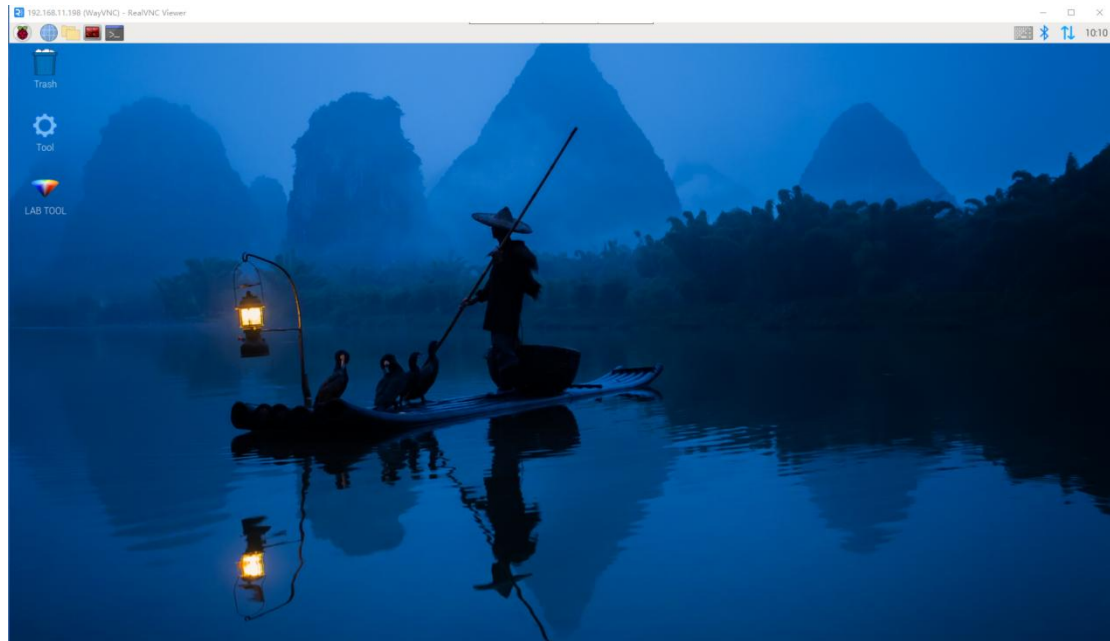
3. Point Cloud

3.1 Depth Camera Point Cloud

 This step is performed on the robot, and when entering commands, it's important to strictly differentiate between uppercase and lowercase letters. You can

also use the "Tab" key to autocomplete keywords.

- 1) Start the robot and connect to the system desktop remotely using VNC.



- 2) Click-on  to open the command-line terminal.

- 3) Execute the below command and hit Enter key to disable the app auto-start service.

```
~/./stop_sh
```



- 4) Open a new command-line terminal, and run the following command to open the depth camera service.

```
ros2 launch peripherals depth_camera.launch.py
```



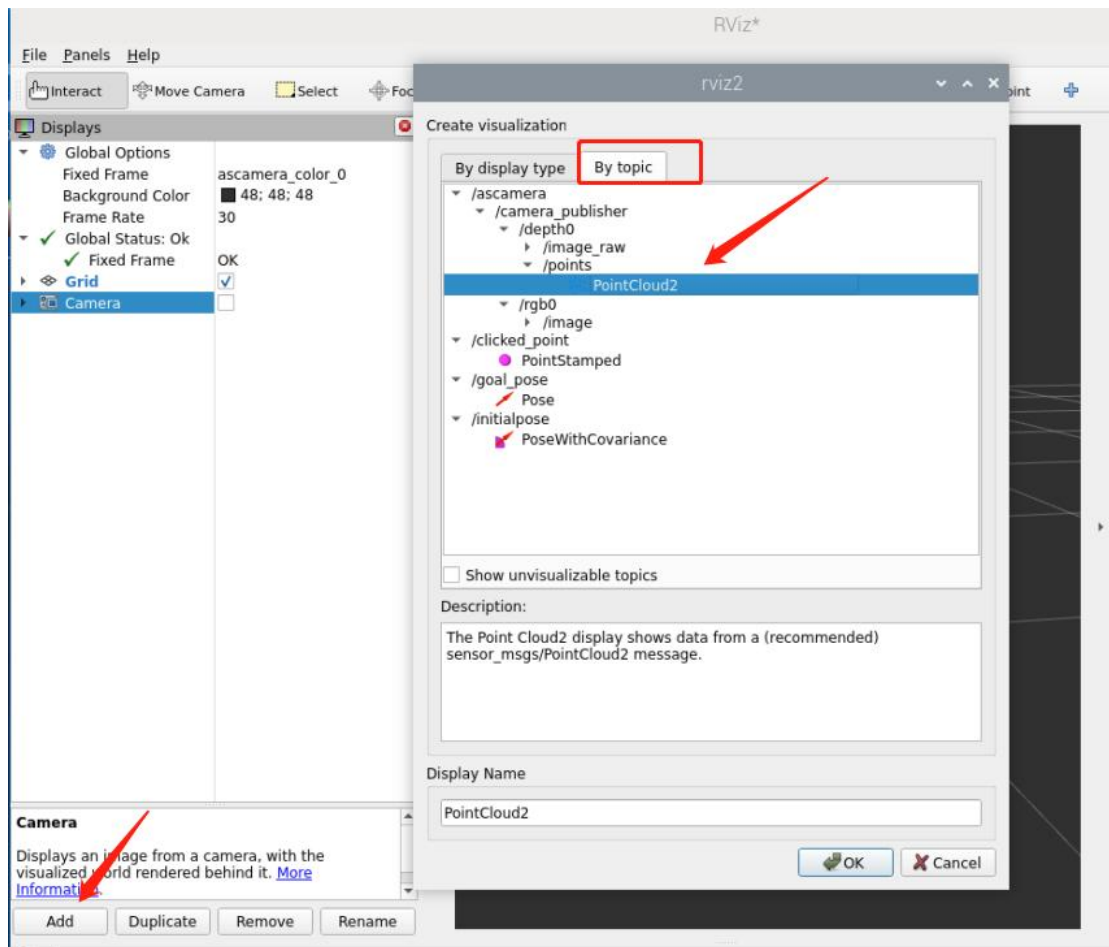
- 5) Open a new command-line terminal and execute the below command to initiate rviz.

```
rviz2
```

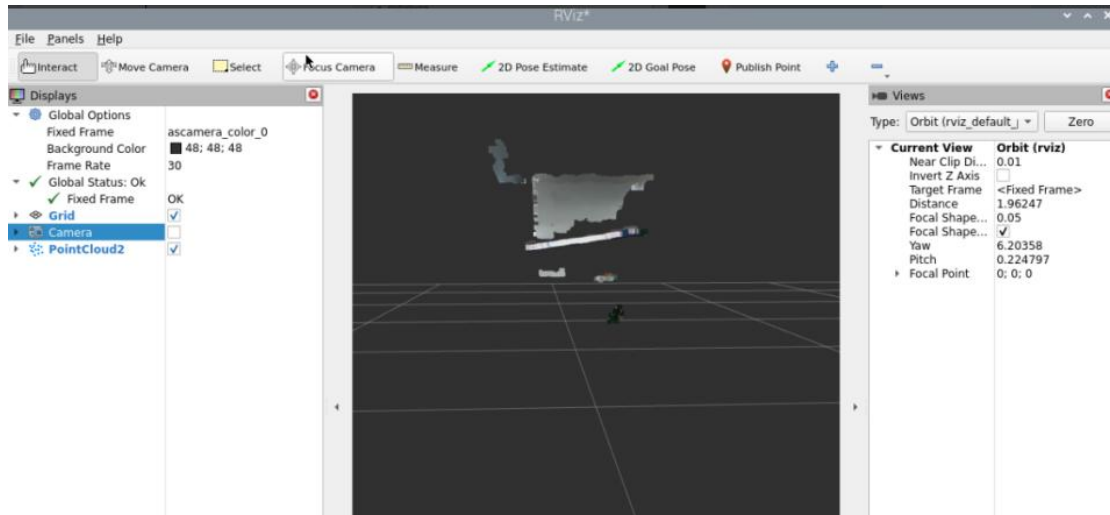
- 6) Choose "ascamera_color_0".



7) Click on "Add" as shown below, then select "By topic" -> "ascamera" -> "ascamera_publisher" -> "depth0" -> "points" -> "pointCloud2", and finally click "OK".



Once you've completed the previous settings, the point cloud will become visible.



8) If you need to terminate the program, press short-cut “Ctrl+C” on the terminal.

After experiencing the Lidar game, you can activate the app service either by using a command or restarting the robot. If the app service is not activated, related app functions will be disabled. In the case of a robot restart, the app service will start automatically.



Click  and enter the command. Press enter to start the app, and wait for the buzzer to beep.

sudo systemctl restart start_node.service

